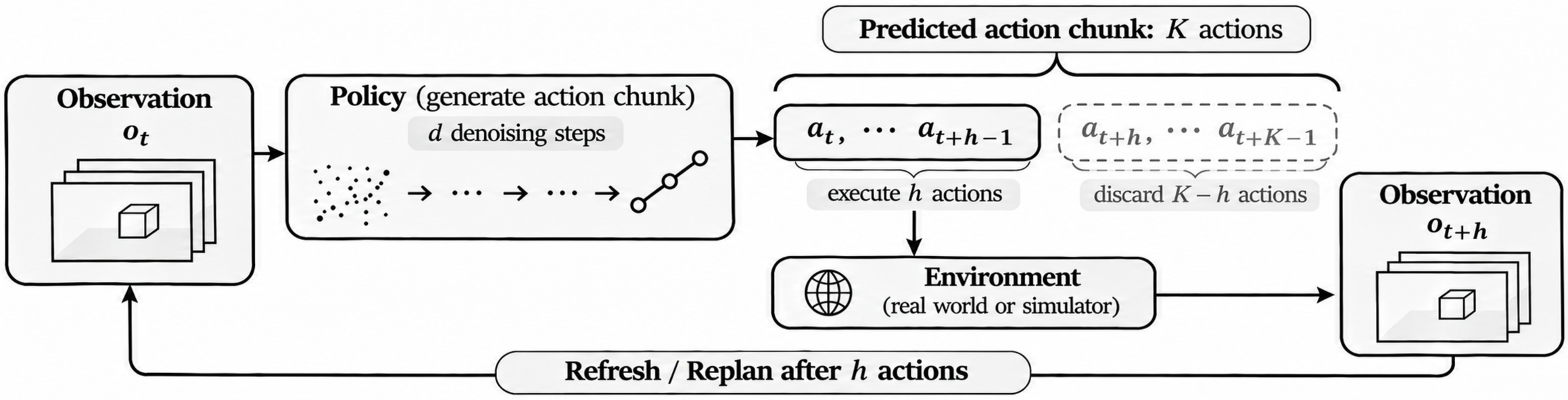
# Refinement vs. Refresh

Compute Allocation for Closed-Loop Robot Policies

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## Chunked Closed-Loop Policy




**Parameters** | $K$ : number of predicted actions per chunk (chunk length) | $h$ : executed actions per replan ($h \leq K$) | $d$ : denoising steps (refinement)

***Synchronous closed-loop execution.*** Each policy call predicts $K$ actions; only the first $h$ are executed before obtaining a new observation and replanning.

**Discovery**
480 attempts · 12 settings
4 Spatial tasks

→ **Freeze**
3 denoising comparisons
before confirmation

→ **Confirmation**
1,120 attempts · 7 settings
8 tasks excluded from selection

Prefix contrasts: later-amended descriptive analyses

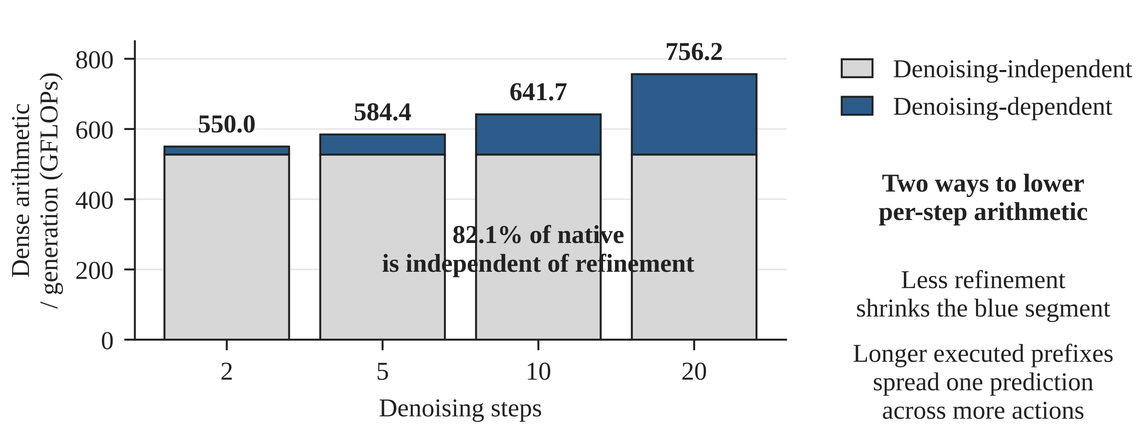
**Setup (held-out)**

**SmolVLA** (frozen) — 50 predicted actions; 2 or 5 denoising steps

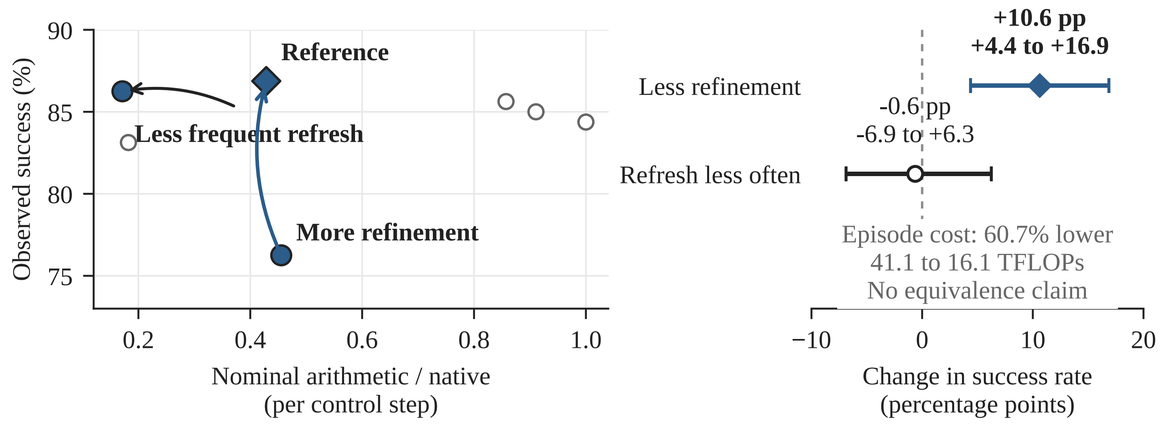
**LIBERO** (simulation) — Refresh after 2 or 5 actions; 8 tasks, 1,120 attempts

**?** Under a fixed inference budget, should we allocate compute to additional denoising steps (**refinement**) or to more frequent observation-conditioned replanning (**refresh**)?

## Why the Trade-Off Exists




## Held-Out Success and Compute




## A Recipe for Closed-Loop Compute Studies

### Evaluate allocations within each model

1. Define refinement and refresh controls.
2. Measure the complete cost of a fresh prediction.
3. Freeze primary comparisons before held-out evaluation.
4. Report success and expenditure together.

### Interpretation boundary

SmolVLA is the worked example, not a universal allocation rule. Arithmetic is not deployment latency, and similar observed success does not establish equivalent quality.